

DEBRA HOGUE, PhD

Computer Scientist | Human-AI Trust & Anomaly Detection Research

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RESEARCH SUMMARY

Computer scientist specializing in human-AI trust calibration and anomaly detection in high-stakes, safety-critical sensor data. Recent work applies unsupervised anomaly detection (autoencoders, Isolation Forest) and topological data analysis (persistent homology) to time-series sensor signals from complex mechanical systems, with direct attention to how detection results are communicated to human operators who must act on them under uncertainty. PhD research introduced a novel framework (MICA) and an empirically validated model (the "Crisis Adoption Problem") for understanding when and why human operators trust or distrust AI-assisted detection systems.

EDUCATION

Ph.D., Computer Science — University of Oklahoma

2026 | GPA: 4.0

Dissertation: Introduced the "Crisis Adoption Problem" and the MICA (Mixed-Initiative Camouflage Analysis) framework, examining human-AI trust calibration in camouflaged object detection. Supported by a 150-person empirical study comparing interactive, mixed-initiative AI assistance (MICA) against a passive baseline system (FACE).

M.S., Computer Science — Oklahoma Christian University

2017 | GPA: 4.0

B.S., Computer Science — University of Central Oklahoma

2016 | GPA: 3.6

CURRENT POSITION

Computer Scientist — 76 SWEG / 557th Software Engineering Squadron, Tinker Air Force Base

Jan 2017-present

- Lead technical contributor on AI-assisted anomaly detection for aircraft sustainment data, applying unsupervised machine learning (autoencoders, Isolation Forest) and topological data analysis to time-series sensor signals from mechanical actuation systems (flap/slat deployment cycles).
- Design and validate detection thresholds for safety-critical alert logic, with attention to reproducibility, cross-validated performance evaluation, and honest characterization of method limitations in the absence of complete ground truth.

RELEVANT TECHNICAL SKILLS

- Anomaly detection: autoencoders (PyTorch, GPU-accelerated), Random Forest, Isolation Forest, topological data analysis (persistent homology, Takens embedding, ripser/persim)
- Time-series & sensor data: nested/structured sensor schemas, large-scale parquet-based pipelines, feature engineering from multi-sensor time series
- Deep learning & computer vision: CNN-based object detection (ResNet50, EfficientDet-D7), pairwise ranking networks (RankNet), generative synthetic data (Stable Diffusion, LoRA fine-tuning)
- Statistical methods: experimental design, ANOVA, power analysis, human-subjects research design (150-person empirical study)
- Languages & tools: Python (pandas, NumPy, scikit-learn, PyTorch), SPSS, CUDA/GPU computing
- Human-AI interaction: mixed-initiative system design, trust calibration frameworks, interpretability and operator-facing explanation design

SELECTED AWARDS

- Dissertation Excellence Award, Gallogly College of Engineering, University of Oklahoma (August 2026)
- Best of Session, IEEE DASC — three consecutive years
- Best Poster Honorable Mention, IEEE VIS 2024
- SMART Scholarship Recipient, Cohort 2021 (DoD SMART Scholarship-for-Service Program)

SELECTED PUBLICATIONS

Building Operator Trust Through Mixed-Initiative Camouflage Analysis for Surveillance 2026 • Accepted, Publication Pending

Hogue, D., Elliott, D.S., & Quadri, G.J. • IEEE 45th Digital Avionics Systems Conference (DASC), Orlando, FL

From Transparency to Agency: The MICA Framework for Addressing the Understanding-Trust Gap in AI-Assisted Camouflage Analysis 2026

Doctoral Dissertation • University of Oklahoma, Gallogly College of Engineering
URI: <https://shareok.org/handle/11244/342777>

MICA: Trust-Driven Design Refinements for Camouflaged Object Detection Applications 2025 • Best in Session

AIAA/IEEE DASC 2025 • Montreal, Canada
DOI: [10.1109/DASC66011.2025.11257236](https://doi.org/10.1109/DASC66011.2025.11257236)

Interactive Visual Analysis of Camouflaged Objects 2025

International Symposium on Visual Computing (ISVC)
DOI: [10.1007/978-3-031-77392-1_33](https://doi.org/10.1007/978-3-031-77392-1_33)

MURDOC: Transforming Pixels into Perception for Camouflage Detection 2024 • Best in Session

IEEE 43rd Digital Avionics Systems Conference (DASC)
DOI: [10.1109/DASC62030.2024.10748781](https://doi.org/10.1109/DASC62030.2024.10748781)

Using Informative AI to Understand Camouflaged Object Detection and Segmentation 2023 • Best in Session

IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC)
DOI: [10.1109/DASC58513.2023.10311193](https://doi.org/10.1109/DASC58513.2023.10311193)

RELEVANT PROJECT EXPERIENCE

Anomaly Detection for Aircraft Sustainment Signals (Current)

Applying unsupervised anomaly detection and topological data analysis to multi-sensor time-series data from aircraft mechanical actuation systems, emphasizing detection of subtle shape/trajectory-based faults (e.g., stalls, overshoots, asymmetric actuation) that simple threshold-based methods miss. Includes cross-validated hyperparameter tuning against ground truth and careful attention to reproducibility and method limitations.

MURDOC / MICA Human-AI Trust Study

Designed and executed a 150-person empirical study comparing an interactive, mixed-initiative AI assistance system (MICA) against a passive AI system (FACE) for camouflaged object detection, examining how interactivity and perceived agency affect operator trust and detection performance.